

# Zhiheng Zhang

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github.com/zhiheng-zhang-Mera

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## Research Profile

Computer Science researcher and research engineer focused on **reliable and secure AI-enabled software engineering**. I am building long-horizon coding-agent infrastructure and multi-agent evaluation mechanisms, with particular interest in automated debugging, repair validation, capability boundaries, and quality assurance on real repositories.

## Education

### University of Melbourne

*Master's degree in Computer Science*

Melbourne, Australia

*Expected 2026*

### University of British Columbia

*Bachelor of Science in Computer Science*

Vancouver, Canada

*Completed Summer 2024*

## Selected Research and Engineering Projects

### DS-Hns — *Long-horizon autonomous software-work runtime*

2026–present

github.com/zhiheng-zhang-Mera/DS-Hns

- Building an execution layer for unattended software work around the official DeepSeek Harness interface, with optional capabilities isolated as extensions.
- Implemented ordered/scheduled task queues, hardware-adaptive local concurrency, unified task lifecycle and terminal events, task history, and terminal notifications.
- Designed extension-failure isolation, restart/recovery-oriented execution, and an idempotent Windows installer with explicit optional-plugin flow.
- Using the system as a testbed for **reliable coding agents, persistent execution, recovery, scheduling, and completion verification**.

### Codex Boss — *Multi-agent orchestration and evidence-based acceptance platform*

2026–present

github.com/zhiheng-zhang-Mera/Codex-Boss

- Building a platform for long-running AI workflows in which multiple agents can plan, execute, critique, and adjudicate work while preserving durable task state.
- Developed platform foundations spanning durable state/events, capability boundaries, plugin isolation, knowledge/data lifecycle, dogfooding, and acceptance/verification.
- Investigating how autonomous systems can distinguish genuine task completion from superficially plausible but weak evidence.
- Current semantic-acceptance work deliberately retains unresolved failure cases: a vacuous case is rejected, while a meaningful case still exposes a false negative in the evidence reader rather than being reported as solved.
- Current regression evidence on the relevant branch includes **2,508 unit tests across 215 files**, plus Electron, renderer, and test typechecking.

### Privacy Lens — *Trustworthy-software / privacy project*

- Explored privacy-risk interpretation and user-facing warning mechanisms with attention to bounded interpretation, auditability, evidence boundaries, and trustworthy software behavior.
- Supports broader interests in trustworthy AI, software governance, and high-stakes automated systems.

### Quant-Ultra — *Empirical quantitative-research and decision-system project*

2026–present

github.com/zhiheng-zhang-Mera/Quant-ultra

- Developed a personal quantitative-research platform with emphasis on robustness, temporal evaluation, data integrity, risk-aware decision making, and reproducible workflows.
- Shifted the project away from headline-return claims toward evaluation quality, non-stationary data, temporal leakage, walk-forward reasoning, and auditable experiment infrastructure.

## Research Interests

AI-enabled software engineering; reliable coding agents; automated debugging and repair validation; software testing and QA for agentic systems; agent security and capability boundaries; empirical evaluation on real repositories.

## Technical and Research Strengths

- End-to-end research engineering: problem formulation, system design, iterative implementation, testing, debugging, and empirical evaluation.
- Multi-module software design with explicit persistence, lifecycle, extensibility, isolation, recovery, and acceptance

criteria.

- Git/GitHub-based experimental development, branch-based iteration, regression testing, and reproducible project documentation.
- Practical multi-AI workflow design: using multiple AI systems as coding/research collaborators while independently reviewing plans, evidence, failure modes, and completion claims.

## Current Research Direction

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I want to study **reliable and secure coding agents for repository-scale tasks**: how execution/recovery infrastructure, testing and debugging signals, capability boundaries, and evidence-based acceptance can jointly reduce silent failures and superficial task completion.